

Chapter 10: Systems of Equations and Inequalities

Name: _____

Date: _____

Instructions: Answer all questions and show clear work. Give exact values when possible; otherwise round numerical answers to the nearest hundredth. Matrix work should display each row operation. Graph any required regions neatly and label all vertices used in optimization.

1. A system of two equations is given. For each, (i) state whether the system is linear or nonlinear, and (ii) find all solutions. (10)

(a)
$$\begin{cases} 2x - y = 5 \\ 3x + 4y = 1 \end{cases}$$

(b)
$$\begin{cases} x^2 + y = 10 \\ x - y = 1 \end{cases}$$

2. Consider the system depending on the parameter k : (10)

$$\begin{cases} x + 2y = 4 \\ 2x + ky = 9 \end{cases}$$

Classify for which values of k the system has (i) no solution, (ii) infinitely many solutions, and (iii) a unique solution. When a unique solution exists, find it in terms of k .

3. Classify each matrix as reduced row-echelon form (RREF), row-echelon form but not reduced (REF), or neither. Justify briefly. (10)

$$(a) \begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (b) \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

4. Use Gaussian elimination to find the complete solution of the system, or show that no solution exists. (14)

$$\begin{cases} x + 2y - z = 4 \\ 2x - y + 3z = 7 \\ 3x + y + 2z = 9 \end{cases}$$

5. Use Gauss–Jordan elimination to find the solution of the system. (12)

$$\begin{cases} x - 2y + z = 0 \\ 2x + 3y - z = 5 \\ 3x - y + 2z = 6 \end{cases}$$

6. At a market, small, medium, and large gift boxes contain the same three items in different quantities. A small box has 1 tea, 2 cookies, and 1 jam and costs \$7.50. A medium has 2 tea, 3 cookies, and 1 jam and costs \$11.00. A large has 3 tea, 4 cookies, and 2 jam and costs \$16.50. Find the individual prices of one tea, one cookie, and one jam. (10)

7. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \\ -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 4 & 5 \end{bmatrix}$. (12)

(a) Explain why $A + B$ is not defined and compute AB .

(b) Verify directly that $(AB)^T = B^T A^T$ by computing both sides.

(c) Compute $A^T A$ and determine whether it is invertible, justifying your answer.

(d) If A is invertible, must A^T be invertible? If so, express $(A^T)^{-1}$ in terms of A^{-1} .

8. Only one of the following matrices is invertible. Use determinants to identify which one, and then find its inverse. (10)

$$A = \begin{bmatrix} 3 & 5 \\ 6 & 10 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 \\ 5 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 8 \\ -1 & -2 \end{bmatrix}.$$

9. Write the system as $A\mathbf{x} = \mathbf{b}$ and solve for (x, y, z) . (10)

$$\begin{cases} 2x + y - z = 1 \\ -x + 3y + 2z = 12 \\ 3x - 2y + 4z = 5 \end{cases}$$

10. Use Cramer's Rule to solve the system, or explain why Cramer's Rule does not apply. (12)
Then compute $\det(A)$, where A is the coefficient matrix, and interpret your result with respect to the solution.

$$\begin{cases} 2x - z = 7 \\ x + 3y = -2 \\ 4x + y + 2z = 10 \end{cases}$$

11. (a) Find constants A and B such that $\frac{8x+1}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}$. (10)

(b) Decompose $\frac{5x-7}{(x-1)(x^2+4)}$ in the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$.

12. Solve each nonlinear system.

(14)

$$(a) \begin{cases} x^2 + y^2 = 20 \\ y = 2 - x \end{cases}$$

$$(b) \begin{cases} x + y = 5 \\ xy = 6 \end{cases}$$

$$(c) \begin{cases} x^2 - xy = 14 \\ x + y = 1 \end{cases}$$

13. Graph the solution set of the system of linear inequalities. Shade the feasible region and label all corner points. (12)

$$(a) \begin{cases} 2x + y \leq 8 \\ x - y \geq -2 \\ x \geq 0, y \geq 0 \end{cases}$$

$$(b) \begin{cases} x + 2y \geq 4 \\ 3x + y \leq 12 \\ x \geq 0, y \geq 0 \end{cases}$$

14. A workshop makes chairs and stools. Each chair requires 4 hours of carpentry and 2 hours of finishing, and yields a profit of \$30. Each stool requires 2 hours of carpentry and 1 hour of finishing, and yields a profit of \$16. The shop has at most 40 hours of carpentry time and 18 hours of finishing time available this week. Let x be the number of chairs and y the number of stools. (14)
- (a) Set up the linear programming model: objective function and constraints, including nonnegativity.
- (b) Graph the feasible region, label its vertices, and determine the production plan that maximizes profit. State the maximum profit.
- (c) Sensitivity: If finishing time increases to 22 hours while carpentry is unchanged, recompute the optimal plan and profit.